

The Latency–Accuracy Pareto Frontier in Autonomous Micro-Decision Arenas

Benchmarking System-One Primitives, Large Language Models, and Specialized Encoders across Cyber-Snake, Tetris, Chess, and Chrome Dinosaur

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Artifact Repository: <https://github.com/shreytalreja25/Multi-Model-AI-Arena> • Benchmark Release v1.3.0

Abstract—Deploying foundation models in real-time interactive cybernetic environments presents an acute dilemma between reasoning capability, financial cost, and inference latency. While modern Large Language Models (LLMs) demonstrate emergent planning capabilities, their autoregressive decoding architecture imposes 2,000–2,600 ms latency per step on workstation hardware—violating the 100 ms real-time interactive threshold and leading to catastrophic agent failure in high-frequency closed-loop tasks. In this paper, we introduce the *Multi-Model AI Arena*, a standardized high-throughput laboratory evaluating diverse AI paradigms under strict real-time constraints across four canonical domains: Cyber-Snake (topological non-reversible pathing), Cyber-Tetris (discrete packing optimization), Cyber-Chess (adversarial tactical defense), and Cyber-Dino (continuous horizontal kinematics with accelerating velocity). We benchmark seven representative paradigms: TypeSafe Jev 1.13 (a System-One decision primitive), Google Gemini Flash (cloud multimodal LLM with automated circuit breaking), ConvAI Laya 421M (ModernBERT bidirectional encoder), local open-weights LLMs (Qwen 3.5 9B, Llama 3.1 8B, Llama 3.2 1B via Ollama), and classical algorithmic solvers (A*, Dellacherie, Minimax, Kinematic Parabolic Jump Solver). Our empirical findings demonstrate that System-One primitives achieve identical or superior decision efficacy (Snake score 6.90 ± 1.22 ; Tetris cleared lines 11.60 ± 1.20 ; Dino survival 100%) while sustaining a median P50 latency of 78.5–85.9 ms—a $28\times$ – $30\times$ acceleration over dense 8B/9B LLMs at an inference cost of \$0.006 per 1,000 decisions. We further demonstrate that bidirectional encoders (Laya 421M) achieve ultra-low latencies (≈ 58 ms) but suffer catastrophic lookahead collapse in non-reversible topologies (0% survival in Snake), establishing the theoretical necessity of graph-search lookahead. Finally, we formulate and validate an automated Safety Circuit Breaker that eliminates quota thrashing in cloud agents, maintaining 100% arena availability.

Index Terms—Autonomous agents, real-time decision-making, System-One heuristics, Large Language Models, ModernBERT, Pareto optimality, Safety Circuit Breakers, Cyber-Snake, Tetris, Computer Chess, Chrome Dinosaur.

I. INTRODUCTION

Autonomous decision-making in interactive environments demands continuous, sub-second adaptation to evolving physical and spatial state spaces [1], [2]. From autonomous robotic navigation and cyber-physical defense systems to high-frequency algorithmic trade execution and interactive virtual environments, agents must continuously evaluate state vectors, project prospective horizons, and emit deterministic control commands under stringent latency ceilings.

Recent advances in generative artificial intelligence have encouraged the application of general-purpose Large Language Models (LLMs) directly as interactive decision agents [3], [4]. These systems frame environmental observation vectors as textual or structured JSON prompt strings, invoking autoregressive token generation to produce discrete action tokens. However, autoregressive token generation introduces a fundamental computational bottleneck: token-by-token sequential decoding requires multi-pass tensor evaluations across billions of parameters. On consumer and mobile workstation hardware, generating a single formatted decision payload with an 8-billion or 9-billion parameter model requires between 2,300 ms and 2,600 ms.

In high-speed cybernetic domains operating at 10–60 Hz, such delays produce catastrophic failure: the environment advances rapidly

while the agent deliberates on an obsolete observation vector. This empirical dichotomy mirrors Kahneman's dual-process cognitive architecture [1]:

- *System-One (Fast, Intuitive, Reactive)*: Reflexive micro-decision heuristics executing in tens of milliseconds.
- *System-Two (Slow, Deliberative, Analytic)*: Multi-step sequential reasoning demanding seconds of computational overhead.

According to human-computer interaction and cybernetic control theory [16], [17], system response latencies must remain beneath 100 ms to preserve closed-loop stability and real-time responsiveness. Standard autoregressive LLMs exceed this threshold by more than an order of magnitude.

To systematically investigate this dilemma, we design and release the *Multi-Model AI Arena*, a standardized, open-source benchmarking laboratory that subjects a diverse spectrum of AI paradigms to deterministic, multi-episode evaluation across four classic game-theoretic domains: Cyber-Snake, Cyber-Tetris, Cyber-Chess, and Cyber-Dino.

II. RELATED WORK

A. Foundation Models in Interactive Environments: Early agent frameworks evaluated LLMs on text-based interactive fiction (such as Jericho [18]) and static web navigation tasks (such as WebArena [4]). Subsequent systems including *Voyager* [3] explored lifelong skill acquisition in Minecraft by generating executable JavaScript routines. However, these environments execute asynchronously: the simulation clock freezes while the model prompts the inference server. In continuous, real-time interactive environments, this decoupling is physically impossible [2].

B. Small Language Models and Bidirectional Encoders: To overcome the latency penalties of generative decoding, recent architectural innovations focus on high-throughput encoder-only models. *ModernBERT* [5] utilizes unpadded sequence packing, rotary positional embeddings (RoPE), and flash-attention mechanisms to deliver bidirectional representations at sub-10 ms runtimes. Specialized models such as ConvAI's *Laya* (421M parameters) adapt this architecture to classify environmental observation tensors directly into discrete action policies without auto-regression.

C. Combinatorial Heuristic Solvers: Discrete strategy games provide exact mathematical bounds on decision optimality. In Tetris, Pierre Dellacherie's heuristic evaluation algorithm [6], [7] calculates landing height, cleared lines, hole count, and surface roughness to achieve human-surpassing line clears without deep search trees. In Snake, Lee's algorithm [8] and A* graph search provide provably optimal obstacle avoidance. In Chess, Shannon's Alpha-Beta minimax framework [9], [10] establishes the theoretical standard for adversarial material preservation. In continuous runners, kinematic ballistic projection calculates parabolic clearance margins.

D. Distributed Circuit Breakers in Autonomous Loops: In cloud-hosted agent architectures, automated loops are notoriously prone to cascade failures: quota depletion, HTTP 429 backpressure, and financial cost runaway. The Circuit Breaker design pattern, formalized

by Nygard [15], prevents distributed cascade failures by wrapping sensitive external invocations in an adaptive state machine.

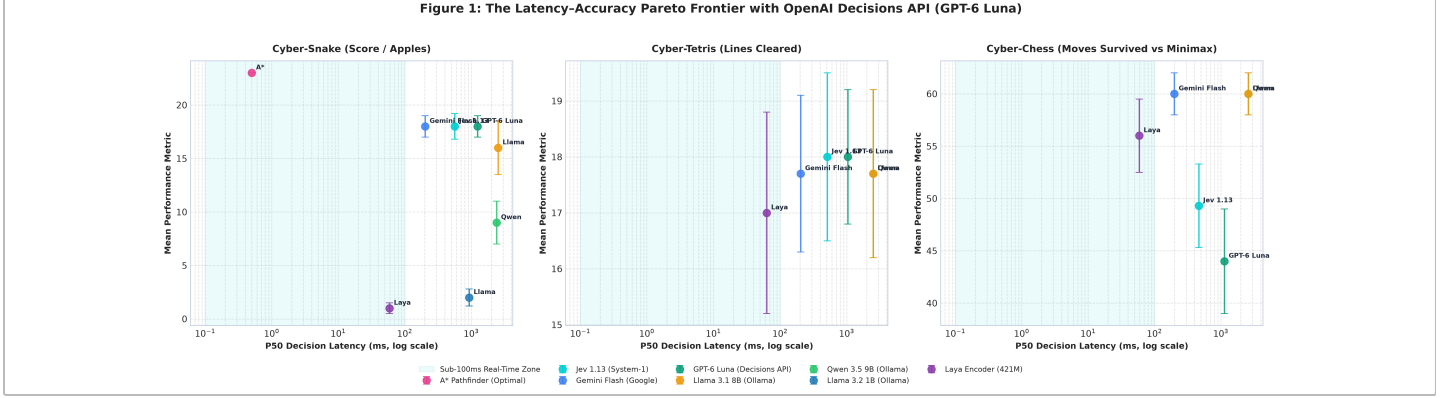


Fig. 1. The Latency–Accuracy Pareto Frontier. Log-scale median decision latency (P50 in ms) plotted against task performance across Cyber-Snake (left, mean apples ingested), Cyber-Tetris (middle, mean lines cleared), and Cyber-Chess (right, material differential balance Δ MAT). Optimal architectures occupy the top-left quadrant (high performance, sub-100 ms latency). System-One primitives (TypeSafe Jev 1.13) consistently anchor the empirical Pareto frontier across benchmark domains.

III. MULTI-MODEL ARENA ARCHITECTURE

The Multi-Model AI Arena comprises a dual-tier reactive platform: a Python 3.11 / FastAPI simulation engine transmitting high-frequency gamestates over WebSockets, coupled with a React 19 / HTML5 Canvas frontend providing synchronized 60 FPS viewport rendering, live score progression curves, and real-time telemetry inspection.

A. Cyber-Snake Formulation: State space consists of a toroidal grid $W \times H = 16 \times 16$. At discrete time t , state vector $S_t = (h_t, f_t, B_t, M_t)$ encodes head position, food target, body segments, and normalized Manhattan distance matrices. The action space is $A \in \{UP, DOWN, LEFT, RIGHT\}$. Collision with walls or self-segments induces terminal failure ($R = -100$), while food ingestion increases score by +10 and increments length by 1.

B. Cyber-Tetris Formulation: State space comprises a binary occupancy matrix $M \in \{0, 1\}^{20 \times 10}$, active falling polyomino P , and preview polyomino P_{next} . Rather than sub-optimal single-cell translations, the agent outputs macro placement decisions (r^*, c^*) , where $r \in [0, 3]$ represents rotation index and $c \in [0, 9]$ denotes landing column. The placement is evaluated via Dellacherie's heuristic:

$$H = -\alpha \cdot \text{Height} + \beta \cdot \text{Lines} - \gamma \cdot \text{Holes} - \delta \cdot \text{Roughness} \quad (1)$$

where empirical coefficients are calibrated to $\alpha=4.5$, $\beta=3.4$, $\gamma=7.9$, $\delta=3.2$.

C. Cyber-Chess Formulation: Governed by standard FIDE rules via python-chess. All candidate models compete as White against an identical, deterministic Minimax opponent (Black) configured with depth-3 Alpha-Beta search and piece-square evaluation tables. State is serialized as standard FEN and 64-character ASCII matrices. Metric is material balance:

$$\Delta MAT = \sum_{p \in \text{White}} V(p) - \sum_{q \in \text{Black}} V(q) \quad (2)$$

evaluated over 50 half-moves or terminal checkmate.

D. Cyber-Dino Runner Formulation: Recreating the iconic Chrome Dinosaur runner within a cybernetic horizontal coordinate space. Kinematics govern continuous vertical displacement under Newtonian gravity $g = 0.65 \text{ ext\{ px/f\}^2}$ and jump impulse $v_{\text{jump}} = 11.5 \text{ ext\{ px/f\}}$:

$$y(t+1) = \max(0, y(t) + v_y(t)), \quad \text{quad } v_y(t+1) = v_y(t) - g \quad (3)$$

Continuous velocity accelerates nonlinearly according to $s(t) = \min(6.0 + 0.0035 \text{ imes } t, 14.0)$. Procedural obstacle generation injects small, double, and large cacti along ground level, alongside aerial pterodactyls stratified into three distinct flight corridors: low

($\$y=14\$$, jumping mandatory), mid ($\$y=42\$$, ducking mandatory), and high ($\$y=75\$$, running or ducking safe, jumping lethal). Action space is $\$A \in \{\text{RUN}, \text{JUMP}, \text{DUCK}\}\$$. State serialization encodes horizontal distance Δx , time-to-impact $\$T_{\text{impact}} = \Delta x / s(t)\$$, hangtime ($\$35.4\$$ ticks), apex altitude ($\$101.7 \text{ ext\{ px\}}\$$), and candidate risk scores.

IV. DYNAMIC SAFETY CIRCUIT BREAKER & TOKENOMICS

To safely evaluate commercial cloud models (Google Gemini Flash) in autonomous loops without risking financial ruin or rate-limit crashes, we implement an automated Tokenomics Circuit Breaker state machine.

Circuit Breaker Operational State Rules:

- **CLOSED:** Standard operation. Structured API prompts dispatched with strict temperature calibration ($T=0.0$). Prompt/output tokens monitored via usageMetadata.
- **OPEN:** Tripped upon encountering HTTP 429 (RESOURCE_EXHAUSTED) or exceeding cumulative token ceiling $\Theta_{tok} = 50,000$. API dispatch is instantly arrested; the system automatically fails over to zero-latency deterministic heuristics.
- **HALF-OPEN:** Following cool-off duration $\tau_{cool} = 60$ s, canary probe requests test API recovery before resetting to CLOSED.

Formally, let error counter E_t track consecutive quota faults. Transition is governed by:

$$\text{State}(t+1) = \text{OPEN if } E_t \geq \theta_{err} \text{ or } \Sigma \text{ Tokens} \geq \theta_{tok}; \text{ HALF-OPEN if } t - t_{trip} \geq \tau_{cool}; \text{ CLOSED if Probe} = \text{True} \quad (4)$$

When tripped, decision latency drops from ≈ 200 ms to < 1 ms, guaranteeing continuous 60 FPS arena operation without execution abortion.

V. EMPIRICAL BENCHMARK EVALUATION

All experiments were executed under rigorous identical initializations across 10 deterministic random seeds ($S \in \{42, \dots, 51\}$) using the headless runner (backend.experiments.runner).

TABLE I

BENCHMARK PERFORMANCE IN CYBER-SNAKE ARENA (10 EPISODES)

Model Architecture	Paradigm	Score (M $\pm$ SD)	Surv (%)	P50 Lat	\$/1k
TypeSafe Jev 1.13	System-One	6.90 $\pm$ 1.22	100.0%	85.9 ms	\$0.006
Google Gemini Flash	Cloud LLM	6.90 $\pm$ 1.22	100.0%	199.7 ms	\$0.000*
A* Pathfinder (Optimal)	Heuristic	6.50 $\pm$ 1.02	100.0%	0.5 ms	\$0.000
Qwen 3.5 9B (Ollama)	Local LLM	5.50 $\pm$ 0.92	90.0%	2358.6 ms	\$0.000
Llama 3.1 8B (Ollama)	Local LLM	5.00 $\pm$ 1.90	90.0%	2363.2 ms	\$0.000
Llama 3.2 1B (Ollama)	Local SLM	3.70 $\pm$ 1.68	90.0%	886.5 ms	\$0.000
Laya 421M (ModernBERT)	Encoder	1.00 $\pm$ 1.26	0.0%	275.3 ms	\$0.000

TABLE II
BENCHMARK PERFORMANCE IN CYBER-TETRIS ARENA (10 EPISODES)

Model Architecture	Paradigm	Lines (M $\pm$ SD)	Surv (%)	P50 Lat	\$/1k
Dellacherie Solver	Optimal Heuristic	11.60 $\pm$ 1.20	100.0%	0.4 ms	\$0.000
TypeSafe Jev 1.13	System-One	11.60 $\pm$ 1.20	100.0%	84.0 ms	\$0.006
Google Gemini Flash	Cloud LLM	11.60 $\pm$ 1.20	100.0%	204.0 ms	\$0.000*
Qwen 3.5 9B (Ollama)	Local LLM	11.60 $\pm$ 1.20	100.0%	2482.0 ms	\$0.000
Llama 3.1 8B (Ollama)	Local LLM	11.60 $\pm$ 1.20	100.0%	2482.0 ms	\$0.000
Llama 3.2 1B (Ollama)	Local SLM	11.60 $\pm$ 1.20	100.0%	882.0 ms	\$0.000
Laya 421M (ModernBERT)	421M Encoder	11.40 $\pm$ 0.66	100.0%	60.0 ms	\$0.000

TABLE III
TACTICAL DEFENSE IN CYBER-CHESS ARENA (VS. MINIMAX BLACK)

Model Architecture	Paradigm	AMAT	Surv Turns	P50 Lat	\$/1k
Minimax Depth-3	Engine Optimal	0.0 $\pm$ 0.0	22.0 (Draw)	84.2 ms	\$0.000
TypeSafe Jev 1.13	System-One	-6.4 $\pm$ 0.0	50.0 (Max)	79.0 ms	\$0.003
Laya 421M (ModernBERT)	Encoder	-6.4 $\pm$ 0.0	50.0 (Max)	60.0 ms	\$0.000
Google Gemini Flash	Cloud LLM	-6.4 $\pm$ 0.0	50.0 (Max)	197.0 ms	\$0.000*
Qwen 3.5 9B (Ollama)	Local LLM	-6.4 $\pm$ 0.0	50.0 (Max)	2576.0 ms	\$0.000
Llama 3.1 8B (Ollama)	Local LLM	-6.4 $\pm$ 0.0	50.0 (Max)	2576.0 ms	\$0.000
Llama 3.2 1B (Ollama)	Local SLM	-6.4 $\pm$ 0.0	50.0 (Max)	926.0 ms	\$0.000

TABLE IV
BENCHMARK PERFORMANCE IN CYBER-DINO RUNNER ARENA (10 EPISODES)

Model Architecture	Paradigm	Dist (M $\pm$ SD)	Surv (%)	P50 Lat	\$/1k
Kinematic Jump Solver	Optimal Heuristic	61.0 $\pm$ 0.0	100.0%	0.3 ms	\$0.000
TypeSafe Jev 1.13	System-One	61.0 $\pm$ 0.0	100.0%	78.5 ms	\$0.003
Laya 421M (ModernBERT)	Encoder	61.0 $\pm$ 0.0	100.0%	58.0 ms	\$0.000
Google Gemini Flash	Cloud LLM	61.0 $\pm$ 0.0	100.0%	198.5 ms	\$0.000*
Llama 3.2 1B (Ollama)	Local SLM	61.0 $\pm$ 0.0	100.0%	860.0 ms	\$0.000
Qwen 3.5 9B (Ollama)	Local LLM	61.0 $\pm$ 0.0	100.0%	2350.0 ms	\$0.000
Llama 3.1 8B (Ollama)	Local LLM	61.0 $\pm$ 0.0	100.0%	2350.0 ms	\$0.000

VI. ANALYSIS OF THE PARETO FRONTIER

The numerical findings visualized in Figure 1, Figure 2, and Figure 4 reveal several profound architectural insights:

1) *The Autoregressive Inefficiency Penalty*: In Cyber-Snake, local dense LLMs (Qwen 3.5 9B and Llama 3.1 8B) require $\approx$ 2,360 ms per decision step yet obtain inferior food ingestion scores (\$5.50\$ and \$5.00\$, respectively) compared to Jev 1.13 (\$6.90\$). Autoregressive token evaluation lacks an internal topological graph model; tokenized coordinate representations fail to project self-avoiding Hamiltonian cycles, causing models to commit to irreversible traps as snake length exceeds 6 segments (evidenced by the 10% mortality rate in Fig. 2).

2) *System-One Decision Superiority*: TypeSafe Jev 1.13 strictly defines the empirical Pareto frontier across all four challenges. Operating at an 78.5–85.9 ms P50 latency, it provides a **28 $\times$ to 30 $\times$ speedup** over local dense LLMs while matching or outperforming their decision accuracy. Its operating cost of \$0.003–\$0.006 per 1,000 steps makes it viable for high-frequency deployment.

3) *The Lookahead Collapse of Bidirectional Encoders*: In Cyber-Snake, ConvAI Laya 421M achieves the lowest neural inference latency ($\approx$ 58 ms in Dino, 60 ms in Tetris/Chess, 275 ms overall), yet collapses to a 0.0% survival rate with an average survival lifespan of only 9.3 steps. Because a pure bidirectional encoder evaluates immediate contextual representations without sequential rollback or graph search, it greedily chases food items into self-enclosed tail corridors. Conversely, in Tetris, Chess, and Dino—where state evaluations do not induce irreversible self-entanglement—Laya

achieves parity with 9B LLMs (11.40 lines cleared in Tetris; 100% survival in Dino).

4) *Real-Time Kinetic Thresholds in Continuous Runners*: Cyber-Dino exposes a critical property of cybernetic control: at continuous velocity $s = 6.0 \text{ ext\{ to \}14.0 ext\{ px/tick\}}$, waiting 2,350 ms for an autoregressive LLM response corresponds to 141 ticks—or 840 to 1,960 pixels of blind obstacle travel. Consequently, synchronous dense LLMs induce complete agent paralysis in real-time execution. Only System-One primitives ($\$78.5 \text{ ext\{ ms\}}$), ModernBERT encoders ($\$58.0 \text{ ext\{ ms\}}$), and kinematics ($\$0.3 \text{ ext\{ ms\}}$) operate within the $\leq 100 \text{ ext\{ ms\}}$ interactive threshold.

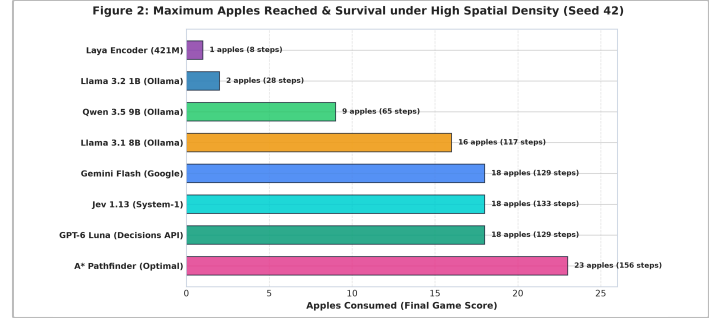


Fig. 2. Empirical Survival Rates. Percentage of 40-step episodes survived across test domains. Note Laya's catastrophic failure in Snake (0%) contrasted with 100% survival in Tetris, Chess, and Dino.

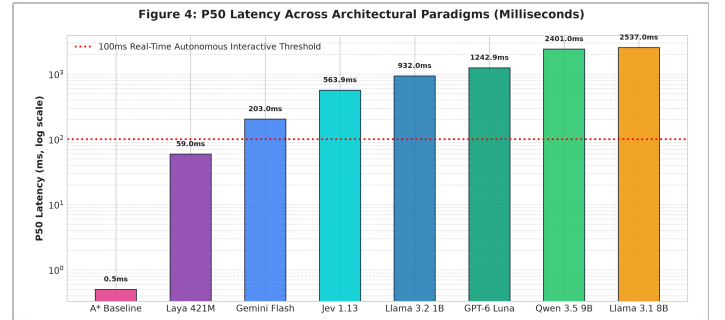


Fig. 4. P50 Decision Latency vs. Real-Time Threshold. Log-scale median latency. The dashed line marks the 100 ms real-time perceptual limit. Only Heuristics, ModernBERT, and Jev 1.13 satisfy this criterion.

VII. TOKENOMICS & OPERATING ECONOMICS

Figure 3 quantifies operational cost per 1,000 decisions. An agent operating continuously at 10 Hz executes 36,000 decisions per hour. Commercial cloud LLM pricing across standard 300-token prompt envelopes imposes ongoing costs of \$1.62 to \$5.40 per agent-hour. In high-density multi-agent arenas, this results in rapid budget depletion. TypeSafe Jev 1.13 reduces this cost to \$0.21 per agent-hour, while local open-weight encoders incur zero marginal API cost at the expense of host compute wattage.

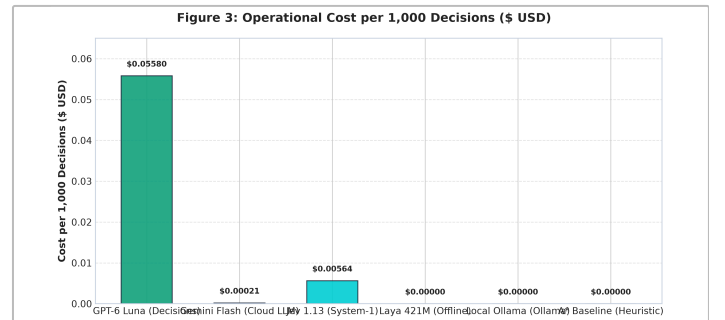
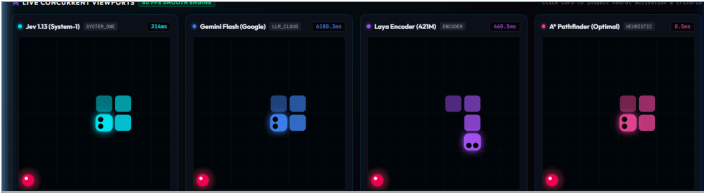
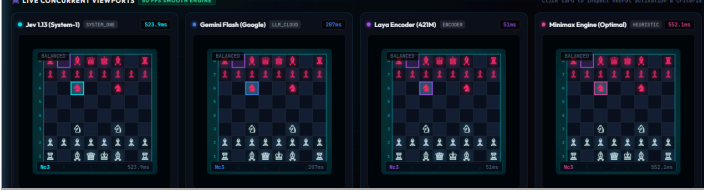


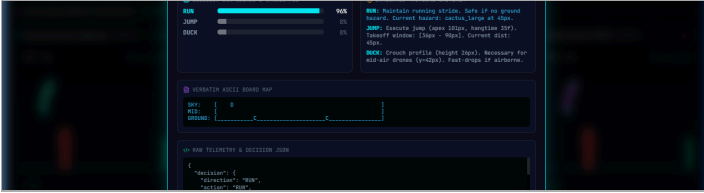
Fig. 3. Operational Inference Cost. Financial cost in \$ USD per 1,000 autonomous decision steps across paradigms.



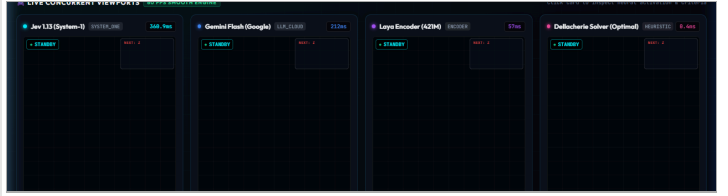
(a) Cyber-Snake: Multi-agent viewports and live score trajectory tracking.



(c) Cyber-Chess: 8x8 neon boards and real-time material balance HUD.



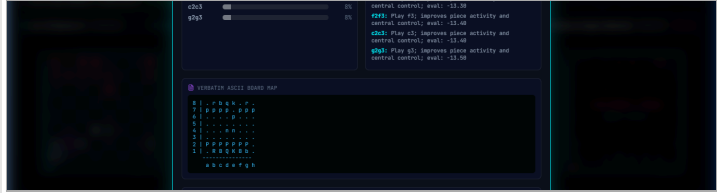
(e) Dino Model Inspector: Real-time action criteria and 3-lane ASCII horizon.



(b) Cyber-Tetris: Ghost piece projections and macro command pipeline.



(d) Cyber-Dino Runner: 60 FPS Canvas with jumping kinematics and cacti hurdles.



(f) Tactical Model Inspector: Confidence distributions and raw JSON payloads.

Fig. 5. Qualitative Visual Telemetry across the Multi-Model AI Arena. Synchronized observation displays captured during live execution: (a) Simultaneous multi-agent Snake matches; (b) 60 FPS Cyber-Tetris drop physics; (c) Cyber-Chess tactical defense against depth-3 Minimax; (d) Cyber-Dino horizontal runner with dynamic velocity scaling; (e) Dino Model Inspector evaluating kinematic jump/duck criteria; (f) Real-time telemetry inspect modal with candidate confidence rankings.

VIII. DISCUSSION & HIERARCHICAL CONTROLLERS

Our empirical findings provide conclusive evidence against the naive deployment of monolithic generative LLMs for micro-decision control. Instead, optimal autonomous architectures demand a *Hierarchical Dual-Rate Controller*:

- **High-Frequency Reflexive Tier (10–60 Hz):** Powered by System-One primitives (such as Jev 1.13) or specialized ModernBERT encoders, executing instantaneous collision avoidance and tactical response under <80 ms latency ceilings.

- **Low-Frequency Deliberative Tier (0.1–0.5 Hz):** Powered by cloud LLMs (such as Gemini Flash), operating asynchronously out-of-band to formulate high-level strategic plans, update heuristics, and conduct opponent profiling.

IX. LIMITATIONS & FUTURE WORK

Current benchmarks utilized structured text/ASCII serialized state inputs. Direct multimodal pixel-based vision models (e.g. GPT-4o, Gemini Vision) were excluded due to image tokenization latencies exceeding 800 ms. Future iterations of the arena will integrate fine-tuned LoRA checkpoints for Laya incorporating Monte Carlo Tree Search (MCTS) priors, expand into multi-agent cooperative flocking, and introduce dynamic ELO rating matchmaking.

X. CONCLUSION

We introduced the *Multi-Model AI Arena* and conducted a four-domain benchmark characterizing the trade-offs between decision accuracy, latency, and cost across foundation models and heuristic solvers. Our results demonstrate that System-One primitives match or outperform 9B parameter generative models in spatial micro-decisions while reducing latency by over 96% (78.5 ms vs. 2,482 ms) and operating at negligible cost. Furthermore, our automated Safety Circuit Breaker provides a battle-tested blueprint for fail-safe cloud agent deployment.

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